



Effect of the logistics industry on the promotion of China's position in the global value chain: An international trade perspective

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ARTICLE INFO

Keywords:

Global value chain position

Logistics industry

Value added of trade

ABSTRACT

In an increasingly complex international environment and deepening economic globalisation, the logistics industry, as the basis of international trade circulation, has a far-reaching effect on promoting a country's position in the global value chain. Based on the perspective of international trade, this study presents a two-way fixed effects model using the 1995–2018 panel data of 59 economies. The empirical results show that the development of the logistics industry can significantly promote the status of a country's global value chain. National economic development level, human capital, technological innovation and application level, and logistics infrastructure level are the main channels through which this promotion effect takes place. Therefore, China should accelerate the construction of export logistics infrastructure and cross-border intelligent logistics systems, and enhance the customs clearance and logistics hub capacity of its ports. 'Reinforcing', 'strengthening', and 'extending' the chain as a means to optimise chain integration force will allow smooth circulation at home and abroad, high-quality development of the logistics industry, and greater opportunities. Ultimately, this will promote the status of the global value chain in the high-quality development stage of the national economy.

1. Introduction

The international trade system is a global value chain (GVC) system in which production links are specialised to promote global economic integration. Inter- and intra-country participation in these GVCs is often unequal. For example, developed countries, Latin America, East Asia, and Southeast Asia show a significant 'smile curve' owing to different regional factors, production stages, and upstream and downstream specialisations (Rosa et al., 2022). In the absence of core technologies, developing countries cannot gain higher competitiveness in the international economy, and risk 'low-end locking', which hinders national economic development.

In recent years, the GVC has become more regionalised. The existing global hubs are centred on China, Europe, and North America, and the cooperation between Europe and North America has a strong influence on China's trade (Sangeeta et al., 2022).

The current international trading system is also undergoing transformation. Smart logistics in China is based on information

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technologies (e.g. 5G technology, the Internet of Things, big data, cloud computing, and artificial intelligence) for improving the level of logistics infrastructure, improving the efficiency of the logistics industry, and reducing operating costs. As part of its hard and soft connectivity policy, China has now begun to focus on countries along the Belt and Road to enhance the resilience of its international logistics supply chain. At home and abroad, it has formed a new logistics industry in cloud logistics, intelligent logistics, cross-border logistics, terminal logistics, etc., as the logistics industry is proving to be an effective means of promoting China's position in the GVC.

Studies on the GVC mostly consider manufacturing and energy; only a few scholars have analysed the logistics industry in this context, but they primarily focus on infrastructure and industrial cluster upgrading. We must then ask: What are the main factors that affect the development of the logistics industry? Can the logistics industry be a core factor affecting a country's position in the GVC? How does logistics development reflect the position of the GVC? The extant literature thus far has not answered these questions.

This study focuses on the question 'Can the logistics industry, as a core factor, affect a country's position in the GVC?' We take a new perspective on how China may enhance its position in the GVC, and our findings should carry implications for policymakers.

The remaining paper is organised as follows: Section 2 reviews the literature and presents the hypotheses. Section 3 develops the research framework and influencing factors based on theoretical analysis. Section 4 presents our method of selecting the variables, introduces the empirical model, and demonstrates the regression analysis. Section 5 concludes.

2. Literature review

2.1. Value-added decomposition theory

The theory of GVC (Hummels et al., 2001) describes the value added process of products, global industrial connection, and industrial upgrading under globalisation. Scholars have decomposed the GVC using various methods, chief among the being those developed by Hummels et al. (2001), abbreviated as the HIY method (Hummels et al., 2001), Wang et al. (2009) or WWP method (Wang et al., 2009), Robert et al. (2011) or KPWW method (Robert et al., 2011), Koopman et al. (2014) or KWW method (Koopman et al., 2014), Wang et al. (2015) or WWZ method (Wang et al., 2015), and the method of Alessandro and Michele (Alessandro, Michele).

Currently, the main indicators to measure the division of labour in the GVC are average transfer step length, upstream degree and downstream degree, number of production stages, export technology complexity index, GVC participation index, and GVC position index. Because the upstream index is affected by the inherent defects in the depth of the production length measurement value, the measured position of the GVC is rough, which makes it difficult to accurately reflect the real position of the GVC. Domestic scholars often use the GVC position index for measurement because its definition is simple, connotation is clear, and direction is clear (Wu, 2019).

Koopman et al. (2012) introduced the GVC participation and status index based on the value-added trade framework. Its basic idea is to compare the indirect value-added exports and foreign value-added of an industry in a country to reflect the international division of labour status of a country in the GVC. The participation index is divided into forward and backward participation indices. The higher the value of the former, the more the country's industry is in the upper reaches of the GVC; the higher the value of the latter, the more it is in the lower reaches of the GVC. The GVC position index is calculated based on the global participation index. The GVC position index reflects the difference between the value of intermediate goods exported by an industry to other countries and the value of the imported intermediate goods included in the exports of this industry.

2.2. Influencing factors of global value chain position

Infrastructure, foreign direct investment (FDI), and industrial agglomeration are considered key factors affecting the promotion of the position of the GVC division of China's manufacturing industry. However, the influence of policies and institutions on the position of the global value division of labour cannot be summarised simply as promoting or inhibiting. The level of infrastructure connectivity promotes the participation rate and depth of forward vertical specialisation of economies in the GVC, and the quality of logistics facilities of trading partners improves the participation degree of Chinese industries in the GVC (Lin & Zhang, 2019; Zhang, 2019). Hao et al. (Hao et al., 2021) believed that infrastructure can improve a country's position in the GVC division. Among these, digital infrastructure has the most significant influence (Li et al., 2020; Qi & Ren, 2021). Digital GVCs have gradually become a driving force for promoting green development (Yaxing & Chi, 2022). FDI can positively influence a country's GVC status upgrading through technological progress and trade network status upgrading (Song & Fang, 2022; Xing et al., 2021; Zhong et al., 2021; Zhu et al., 2022). The stability of GVC can be realised by improving the technology level, optimising the industrial structure, and expanding the trade scale. Industrial agglomeration promotes the status of the GVC in the manufacturing industry (Su et al., 2020; Wang & Du, 2022; Yang & Li, 2018) after companies in the same industry cluster in a certain area.

2.3. Development and influencing factors of the logistics industry

The division of labour in the GVC of China's logistics industry is deepening, and the international logistics market and development scale are increasing, showing a positive trend. However, compared with developed countries, China's logistics industry is still relatively weak in terms of international port shipping business, added value of trade exports, road transport, and vehicle ownership. This has conferred a low status to China's logistics industry in the division of the GVC (Wang & Liu, 2019).

Although the breadth and depth of China's participation in GVCs are not obvious, the advantage of value-cost conversion is the

most prominent (Dong et al., 2019). With the rapid development of the logistics industry, there are problems of high logistics cost, low efficiency, and low degree of specialisation (Lv & Wang, 2017). Under the GVC, the logistics industry aggregation needs to be upgraded, which can be realised through linkages with the manufacturing industry and cooperation with foreign logistics enterprises (Deng & Zhang, 2012).

Currently, there exist three high-quality development paths for China's logistics industry: the traditional path with a hinterland economy as the core condition; an open path with an opening level as the main core condition; and a comprehensive path with innovation, government intervention, and opening as the main core condition, respectively (Cao et al., 2020). Different levels of enterprises are starting to develop 'intelligent + network' to improve logistics efficiency, but there are also shortcomings in upgrading the logistics industry. We can promote the intelligent + network development of the logistics industry in the age of science and technology by utilising reasonable regional layout planning, strengthening the construction of talent teams, and scientific and perfect laws and regulations (Huang & Wang, 2021).

There is a relatively close relationship between the logistics and manufacturing industries (Wang & Zhao, 2021). Technological innovation plays a leading role in the logistics industry's development. The improvement in the technological innovation ability of the logistics industry has promoted the development of the modern service industry (Su & Liu, 2022). The digital economy promotes high-quality development of the logistics industry through technological innovation and improvement of factor allocation efficiency (Luo & Qinmei, 2022). In recent years, the application of artificial intelligence technology in the field of consumption and circulation has improved circulation efficiency and promoted higher-quality development in the logistics industry (Chen, 2021). In addition, environmental regulations (Zheng & Qin, 2021), changes in export trade (Zhan et al., 2019), financial agglomeration and development (Li, 2019), and other factors have affected the logistics industry's development.

2.3.1. Literature review

Literature reviews show that domestic and foreign scholars study the position of the division of labour in GVCs from different perspectives and research objects. However, there is still a lack of further research on the direction of the logistics industry's participation in the division of labour in the GVC.

First, relevant research on China's logistics industry from the perspective of GVC is relatively scarce, and the research years are relatively long, which to some extent is not close to the current status of the national logistics industry participating in the division of labour in the GVC.

Second, there are few quantitative studies on the Chinese logistics industry's participation in GVC, and the database used for measurement is outdated. Hence, the measurement results have weak timeliness.

Third, few scholars provide targeted suggestions based on measuring the distribution status of the logistics industry in GVC. In this study, we use the method of trade added value to measure the GVC, and thus analyse the effect of the logistics industry on the promotion of GVC status at the national level. We then provide policy suggestions based on the main influencing factors and China's actual situation.

3. Research framework

As participation in the GVC increases, we observe that the logistics industry influences the scale, spillover, and competition effects in the GVC system. First, with the improvement in the scale effect, the funds available for product research and development of logistics enterprises increase, enhancing the innovation technology of logistics enterprises. This effect is conducive to improving the efficiency of logistics enterprises and promoting the efficiency of the logistics industry. At the same time, it can significantly reduce the marginal cost of the innovation process and improve the internal power of logistics enterprises' independent innovation.

Second, the number of intermediate services in which logistics enterprises participate increases significantly, and the number of transactions is also increasing in tandem. This phenomenon is conducive to the technological progress of service technology and management knowledge.

Finally, an increase in participation in GVC will inevitably lead to an increase in the number of logistics competitors, an increase in competitive pressures on logistics enterprises, and, ultimately, the promotion of independent innovation by logistics enterprises. As a result, these enterprises can then increase investment in research and development, and thus form their core competitiveness.

The gradual transformation and upgrading of the logistics industry significantly affect the level of national economic development. Normally, the higher the GDP of a country, the higher the expenditure on the logistics industry, which allows for the yearly increase in the absolute value of the total social logistics cost.

The development level is proportional to GDP, and the increase in the added value of logistics is higher than the growth level of GDP. The level of the national logistics infrastructure greatly determines its transportation capacity and has an important influence on the efficiency of international trade. The high cost of the logistics industry is closely related to serious delays in infrastructure and equipment conditions. To some extent, the level of fixed capital and urbanisation restricts the economic level of a country. With the development of science and technology, living conditions improve, higher education enrolment surges, and the resulting educated talents settle in developed cities. Industry talents propel independent innovation of enterprises, creating high-tech products that ultimately promote national economic development. This sequence of events should create positive influence on the promotion of GVC status.

At the same time, anti-globalisation has taken root in the international trade system as well. The trade protection of developed countries shows both diversification and compulsion. These include green barriers, technical barriers, and trade management. However, anti-globalisation movements cannot stop economic globalisation. For instance, the Belt and Road is a new platform for all-

dimensional and multi-tiered international economic cooperation and is also a leading global logistics service provider. It has transformed from providing global logistics services to providing global supply chain services, which are highly integrated with the new generation of information technology. Through it, China will gradually transform and upgrade to an intensive growth model. Furthermore, investments in logistic-related infrastructure, such as 5G, data centres, and artificial intelligence, are increasing. Through deep integration of digital technology and logistics livelihood services, mutual promotion and progress in upgrading consumption and industries will be possible.

Moreover, the logistics supply chain will shift from the traditional mode of simple upstream and downstream division of labour cooperation to cross-border integration of the intelligent development mode. Indeed, the cross-border integration of supply chain logistics distribution and e-commerce has formed a new intelligent logistics ecology of ‘new retail + scene logistics + vehicle and cargo matching + supply chain finance’ (Dong, 2021) (Dong, 2021). By promoting the logistics industry to form wholesale and retail, present and future, and tangible and intangible market development, China’s GVC status can be further strengthened. Based on this theoretical analysis, a basic theoretical framework was constructed, as shown in Fig. 1.

4. Empirical analysis

This study analyses the main factors influencing countries’ participation and position in GVCs. The more significant the influencing factor, the higher the factor index, and the more significantly will a country’s GVC status improve. In this analysis, the main influencing factors common to many countries were modified to reflect the Chinese context, and then applied to our model for improving China’s position in the GVC. Thus, the bidirectional fixed-effects model was used for regression analysis.

Following Alessandro and Michele’s (Alessandro, Michele) method, the GVC participation index is decomposed and the GVC position index is calculated as the explained variable for analysing the influencing factors of GVC position. We measured the GVC and status thereof from four aspects: logistics industry level, national economic development level, technological innovation and application level, and digital economy level. Significance was analysed as explanatory variables.

To appropriately include the main factors affecting the promotion of GVCS in the measurement model, control variables should be added to avoid missing unmeasurable factors that cause an estimation bias in the model. Considering the difficulty of data acquisition, control variables which include material capital level, urbanisation level, human capital level, logistics infrastructure level, and foreign direct investment were selected.

4.1. Model settings, variables, and data source

4.1.1. Model settings

We selected the relevant statistical data of the logistics industry of 59 countries for 24 years, from 1995 to 2018. The explanatory variables were mainly selected from the logistics industry to eliminate the heteroscedasticity of the explanatory variables. In addition, the explanatory variables were logarithmic in order to construct the equation. Using the GVC position index of the logistics industry as the explained variable, the following equation was established:

$$GVC_{ir} = \alpha_0 + \alpha_1 \ln TRAN_{ir} + \alpha_2 X_{ir} + \varepsilon_{ir},$$

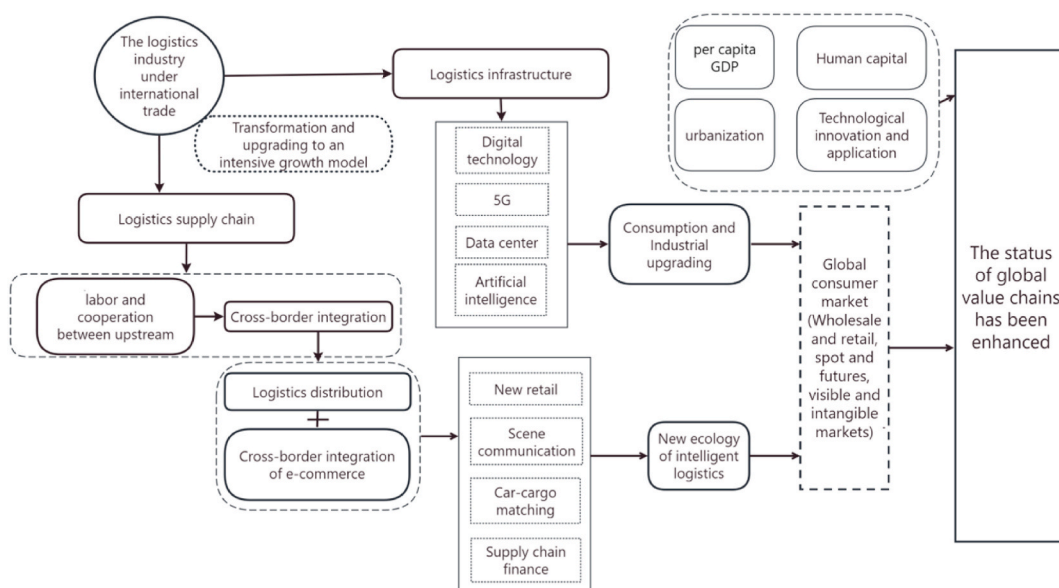


Fig. 1. The effect of logistics on the promotion of global value chains.

where i is the country, r is the year, and GVC_{it} is the GVC position indicator. $TRAN_{it}$ indicate the explanatory variables. X_{it} represents a set of control variables.

4.1.2. Variables and data source

4.1.2.1. Explained variables. Global value chain position indices The global value chain participation index was calculated according to the decomposition method proposed by Alessandro and Michele (Alessandro, Michele). The GVC backward participation index measurement formula is as follows:

$$GVC_backward_{sr} = \frac{V_s(I - A_{ss})^{-1} \sum_{j \neq s}^G A_{sj} B_{js} E_{sr} + \sum_{t \neq s}^G V_t B_{ts} E_{sr}}{u_N E_{sr}}$$

Moreover, the GVC forward participation index measurement formula is as follows:

$$GVC_forward_{sr} = \frac{V_s(I - A_{ss})^{-1} A_{sr} (I - A_{rr})^{-1} \left(\sum_{j \neq r}^G Y_{rj} + \sum_{j \neq r}^G A_{rj} \sum_k^G \sum_{l \neq s}^G B_{jk} Y_{kl} \right)}{u_N E_{sr}}$$

An index of participation in GVCS can be constructed as follows:

$$GVC_{sr} = GVC_backward_{sr} + GVC_forward_{sr}$$

Next, the formula for calculating the global value chain position index is

$$GVC_{position} = \ln(1 + GVC_{forward_{sr}}) - \ln(1 + GVC_{backward_{sr}}),$$

where E_{sr} is the total exports from country s to country r . u_N is the $1 \times N$ unit row vector. V_s is the $1 \times N$ vector containing the value-added share of the total output per unit produced in country s . $(I - A_{ss})^{-1}$ represents the domestic inverse Leontief matrix. $\sum_{j \neq s}^G A_{sj} B_{js} E_{sr}$ is the intermediate product that re-enters the foreign country in the s to r export. $\sum_{t \neq s}^G V_t B_{ts} E_{sr}$ is the foreign content of exports. $(\sum_{j \neq r}^G Y_{rj} + \sum_{j \neq r}^G A_{rj} \sum_k^G \sum_{l \neq s}^G B_{jk} Y_{kl})$ is the r country (re)exports divided into intermediate goods and final goods.

4.1.2.2. Explanatory variable. Logistics industry level. In recent years, the logistics industry has developed rapidly and occupied an important position in the global economy. Whether domestic or international, the level of a country's logistics industry has a significant influence on not only economic development but also foreign exports. For reference (Alessandro, Michele), we selected total exports. Therefore, the D49–D53 total exports of the OECD–TiVA database were selected for the analysis. The total exports of five industries—land transport and pipeline transport, water transport, air transport, transport storage and support activities, and postal and express activities—were taken to represent the level of logistics.

4.1.2.3. Control variables. National economic development. The economic development level of a country has a significant influence on the position of GVC. For reference, we used the per capita GDP of the country.

Technological innovation and application level. Products require constant technological innovation. The higher the level of technology needed to export the product, the higher the value added and the more upstream a country is in the GVC. However, there are challenges to measuring a country's technological innovation level simply by the number of resident patent applications. Technological innovation is mainly applied to actual products to produce economic benefits. Therefore, we chose the export value of high-tech products to represent the level of technological innovation and application.

Human capital. Strides in education have allowed a greater number of people to enter diverse industries, triggering sustainable industrial development, healthy enterprise competition, and improvements in the GVC position. For reference (Xing et al., 2021), we used the enrolment rate of colleges and universities.

Logistics infrastructure level. Infrastructure contributes significantly to the position in GVCs. For reference (Hao et al., 2021), infrastructure was divided into transportation and communication to analyse the role of infrastructure in the GVC. We also expressed the level of logistics infrastructure by the average air freight volume, railway freight volume, and container terminal throughput.

Urbanisation level. Industrial transfer and concentration are also an effect of economic development, and this has increased the size of urban populations. Urbanisation affects the position of GVCs to some extent, and thus, for reference (Xing et al., 2021), we employed the proportion of urban population in total population.

Foreign direct investment. National governments alleviate underinvestment and expand employment by pouring foreign capital into the economy. The technological spillover effect brought about by FDI can be used to realise industrial upgrading and promote the growth of the real economy, prompting us to employ it as one of our control variables.

4.1.2.4. Data source. The original data of GVC position index and logistics industry level were gathered from TiVA (Trade in Value Added, 2021), jointly released by OECD and WTO. Data for the remaining variables were obtained from the World Bank World

Development Index database. As some national indicators are missing, interpolation and extrapolation methods were used to supplement them. Table 1 lists the specific definitions and data sources of each variable.

4.2. Empirical process

4.2.1. Benchmark regression

Stata 17.0 was used for regression analysis. Note that some data pertaining to the variables of *Inno*, *CER*, *INT*, and *FDI* were missing. After linear interpolation, the variables included were *Inn*, *CERR*, *INTS*, and *FDIA*. We took the logarithm of the explanatory variable and the control variable. Table 2 lists the descriptive statistical results of the model variables.

As shown in Table 2, we obtained 1416 observed sample values after data processing. In 59 sample countries, the maximum value of logistics level was 11.92 and the minimum value was 2.398, pointing to a large economic and logistics development gap among countries. After grasping the basic characteristics of the data, we tested for multicollinearity among the variables to ensure the accuracy of the model estimation. Accordingly, we employed the variance inflation factor (VIF) to examine the correlation among variables. Severe multicollinearity does not exist when the test value is less than 10. In our analysis, Stata measured VIF values among the variables, as shown in Table 3. The mean VIF value of each variable was 1.89, and all values were lower than 10, indicating no multicollinearity.

Before the empirical analysis, we used scatter plots and fitting lines to intuitively demonstrate the relationship between the level of the logistics industry and the position of GVC. Figs. 2 and 3 respectively show the linear fitting of the logistics industry level with GVC participation and GVC position. The horizontal axis is the level of the logistics industry. The vertical axis is the corresponding GVC position index. The line sloping to the upper right in Fig. 2 intuitively shows the positive correlation between logistics levels and participation in the GVCS. However, Fig. 3 shows that the level of the logistics industry indicates a gentle improvement in the GVC status.

Fig. 4 shows China's position in the GVC from 1995 to 2018. Although the position is quite low, we observe a gradual upward trend. Figs. 4 and 5 respectively show the changing trends of China's GVC position and logistics industry level. Both show sustained and consistent growth trends. These results simply and intuitively verify the positive correlation between the development level of China's logistics industry and the promotion of its GVC position. These results informed our subsequent empirical regression.

After the above test and simple linear fitting, the sample data of 59 countries from 1995 to 2018 were estimated by regression. The Hausman test confirms the necessity of using the fixed effects model. To improve the readability and practicability of the regression results, the two-way fixed effects model was selected. To improve the robustness of the test results, and to investigate the influence of the core explanatory variable on the explained variable, we incorporated the explanatory variable into the equation and gradually added the control variable. Table 4 shows the basic regression results of the influence of the logistics industry level on GVC position.

In model (1), when the explanatory variables were introduced, the effect of the logistics industry level on GVC position was significant at the level of 0.1%. The estimated coefficient was 1.195. For every 0.1% increase, the GVCS rose by about 0.01195 units. The control variables of national economic development level, technological innovation and application level, human capital, logistics infrastructure level, urbanisation level, and FDI were then added successively in model (2) to model (7). The level of the logistics industry significantly influenced the position of GVC at the level of 0.1%, and its coefficients were 1.241, 1.367, 1.381, 1.378, 1.33, and 1.291, respectively. Thus, the logistics industry was shown to significantly promote the status of GVC.

In terms of control variables, national economic development level and urbanisation level can be significant at the level of 0.1%. Models (3), (6), and (7) show that the level of technological innovation and application can be significant at the 0.1% level, and models (4) and (5) can be significant at the 1% level. Model (7) shows that FDI can be significant at the 1% level. Thus, national economic development level, urbanisation level, technological innovation and application level and foreign direct investment were found to have a positive promoting effect on the promotion of GVC status. However, the level of human capital and logistics infrastructure had no such effect on GVC participation.

Table 1
Variable description and data source.

Type	Variables	Symbols	Indicators	Data source
Explained variables	GVC positionv indices	<i>GVC-position</i>	Relative division position in GVC	RIGVC UIBE; original data from OECD 2021
Explanatory variables	Logistics industry level	<i>TRAN</i>	D49–D53 Total export	TIVA
Control variables	National economic development	<i>PGDP</i>	Per capita GDP of each country	World Bank database
	Technological innovation and application level	<i>Inno</i>	Value of high-tech exports	World Bank database
	Human capital	<i>CER</i>	College enrolment rate	World Bank database
	Logistics infrastructure level	<i>INT</i>	Air freight, rail freight, container terminal throughput	World Bank database
	Urbanisation level	<i>CIT</i>	Urban population as a percentage of total population	World Bank database
	Foreign direct investment	<i>FDI</i>	Foreign direct investment	World Bank database

Table 2

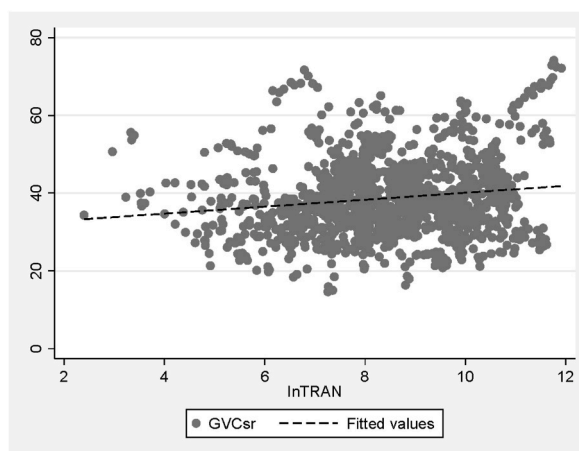
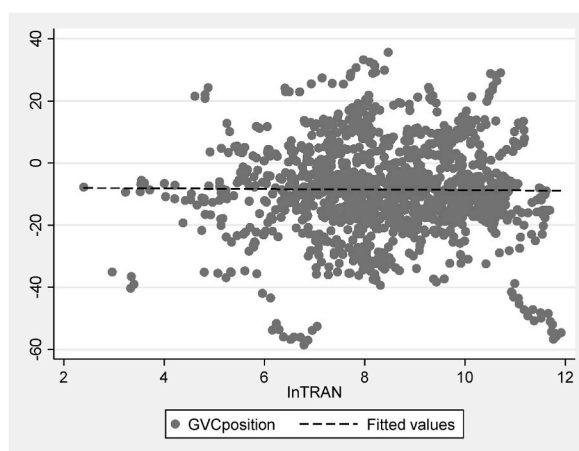
Descriptive statistics of variables.

Variable	N	Mean	SD	Max	Median	Min
<i>GVCposition</i>	1416	−8.576	14.68	35.61	−9.226	−58.53
<i>lnTRAN</i>	1416	8.479	1.636	11.92	8.511	2.398
<i>lnPGDP</i>	1416	9.378	1.288	11.72	9.570	5.497
<i>lnInn</i>	1416	21.92	2.921	27.90	22.24	3.921
<i>lnCERR</i>	1416	3.765	0.698	4.962	3.970	0.235
<i>lnINTS</i>	1416	11.53	3.802	18.18	12.71	−3.202
<i>lnCIT</i>	1416	4.213	0.315	4.605	4.299	2.851
<i>FDIA</i>	1416	−0.00500	0.339	1.773	−0.0100	−3.454

Table 3

Multicollinearity results.

Variable	lnPGDP	lnCERR	lnCIT	lnINTS	lnInn	lnTRAN	FDIA	Mean
VIF	3.060	2.790	2.560	1.360	1.280	1.160	1.040	VIF
1/VIF	0.327	0.359	0.391	0.736	0.781	0.866	0.965	1.890

**Fig. 2.** Fitted values of *GVCsr* and *TRAN*.**Fig. 3.** Fitted values of *GVC position* and *TRAN*.

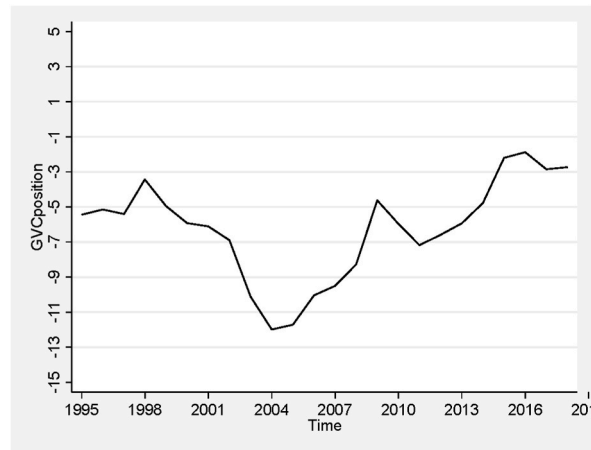


Fig. 4. China's position in the GVC.

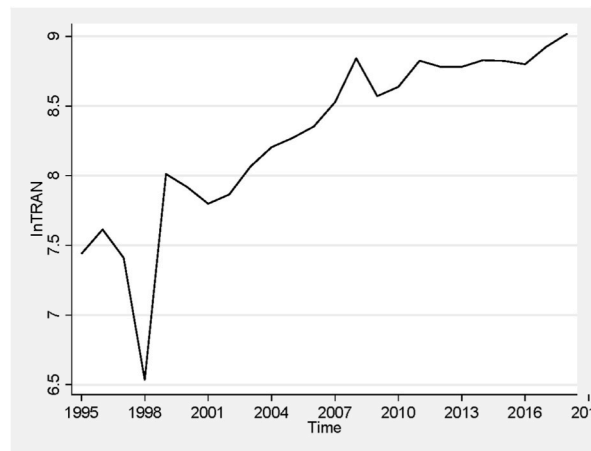


Fig. 5. Level of China's logistics industry.

4.2.2. Heterogeneity test

To explore whether the level of the logistics industry in countries with different levels of economic development had the same effect on the status of GVC, we analysed the heterogeneity of countries at different levels of economic development. The 59 countries are divided into developing and developed countries, and a regression analysis of the GVC position index of two kinds of countries was conducted to explore any heterogeneity in the influence of the logistics industry level on the GVC position. The regression results after classification are shown in Table 5.

The results of sample grouping regression show that the level of the logistics industry in developing countries had a significant effect on the position of GVC at the level of 0.1%. The coefficient was 1.939. However, the level of the logistics industry in developed countries had no significant influence on the position of GVC. Thus, the level of the logistics industry had different effects on the GVC status of countries with different economic development levels. We understand this to mean that developing countries are generally at the lower end of the GVC. Therefore, they are more likely to realise the promotion of GVC status through logistics development. However, developed countries have abundant natural resources put to efficient use or a large number of core technologies, while having transferred most material production. Their financial industry is capital-rich, leading to industrial hollowing. Developed countries are in the upper reaches of the GVC and have little room for improvement.

4.2.3. Endogeneity test

According to the basic regression analysis, the level of the logistics industry plays a significant role in promoting the status of GVC. However, there may be an endogeneity problem between the two from two main sources: On the one hand, the model may miss important explanatory variables, so we employ a fixed effect model with fixed individual effects to solve the problem. On the other hand, the explained variable and the core explanatory variable interact; in other words, the position of the GVC can affect the level of a country's logistics industry in a reverse manner. Therefore, we used the system generalised method of moments (GMM) method to introduce the lag term of the GVC position of the explained variable into the econometric model. The lag period of the explanatory

Table 4
Benchmark regression.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>
<i>lnTRAN</i>	1.195*** (0.000)	1.241*** (0.000)	1.367*** (0.000)	1.381*** (0.000)	1.378*** (0.000)	1.330*** (0.000)	1.291*** (0.000)
<i>lnPGDP</i>		2.734*** (0.000)	2.351*** (0.000)	2.044*** (0.000)	2.042*** (0.000)	1.764*** (0.000)	1.903*** (0.000)
<i>lnInn</i>			0.311*** (0.001)	0.286** (0.002)	0.284** (0.002)	0.381*** (0.000)	0.379*** (0.000)
<i>lnCERR</i>				0.968 (0.072)	0.975 (0.072)	−0.776 (0.205)	−0.798 (0.192)
<i>lnINTS</i>					0.009 (0.903)	0.016 (0.813)	0.009 (0.898)
<i>lnCIT</i>						14.773*** (0.000)	14.852*** (0.000)
<i>FDIA</i>							0.940** (0.009)
_cons	−15.892*** (0.000)	−40.387*** (0.000)	−44.290*** (0.000)	−44.255*** (0.000)	−44.273*** (0.000)	−99.272*** (0.000)	−100.406*** (0.000)
<i>N</i>	1416	1416	1416	1416	1416	1416	1416
adj. <i>R</i> ²	−0.017	0.008	0.016	0.018	0.017	0.041	0.045

p-values in parentheses.

**p* < 0.05.

***p* < 0.01.

****p* < 0.001.

Table 5
Grouping regression results for developing and developed countries.

Variable	Developing country	Developed country
	<i>GVCposition</i>	<i>GVCposition</i>
<i>lnTRAN</i>	1.939*** (0.000)	0.492 (0.104)
<i>lnPGDP</i>	3.407*** (0.000)	−1.395* (0.029)
<i>lnInn</i>	0.677** (0.002)	0.272** (0.003)
<i>lnCERR</i>	−1.847 (0.058)	0.061 (0.933)
<i>lnINTS</i>	−0.090 (0.412)	−0.058 (0.497)
<i>lnCIT</i>	6.268 (0.067)	12.427* (0.016)
<i>FDIA</i>	1.926*** (0.000)	−0.342 (0.497)
_cons	−78.840*** (0.000)	−59.179** (0.009)
<i>N</i>	600	816
adj. <i>R</i> ²	0.190	0.050

p-values in parentheses.

**p* < 0.05.

***p* < 0.01.

****p* < 0.001.

variable of FDI was also used as an instrumental variable to solve the above problem. This method can effectively avoid endogeneity among variables and between variables and residuals. The endogeneity test results of the system GMM method are shown in Table 6.

The Sargan test requires a test value greater than 0.1 to pass, while the Arellano–Bond test requires an AR(1) test value less than 0.1 and an AR(2) test value greater than 0.1. Table 5 shows that the model passed both tests, confirming the correctness of the estimated result.

By comparing the results in Tables 4 and 5, we found that the coefficients of the explanatory variable logistics level were significantly positive, and the resulting difference was only numerical.

Further, by taking the lag period of the explanatory variable to the level of the logistics industry as an instrumental variable, we may avoid any endogeneity to a certain extent. The regression results in Table 5 confirm the reliability of the basic regression results in Table 4.

Table 6
System GMM regression estimation results.

Variable	(1)
	<i>GVCposition</i>
<i>L.GVCposition</i>	0.912*** [0.1019]
<i>lnTRAN</i>	0.237 [0.4994]
<i>lnPGDP</i>	−0.633 [3.9932]
<i>lnInn</i>	0.433 [0.5397]
<i>lnCERR</i>	−3.592 [4.9466]
<i>lnINTS</i>	0.0386 [0.0997]
<i>lnCIT</i>	25.28 [26.1668]
<i>FDIA</i>	−0.106 [0.8280]
_cons	−99.78 [94.0272]
Sargan test	0.015
Arellano–Bond AR(1)	0.000
Arellano–Bond AR(1)	0.014
Observations	1416

Standard errors in brackets.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

4.2.4. Robustness test

To ensure the robustness of the empirical results above, we used the index substitution method (Hao et al., 2021) (Hao et al., 2021). We replaced the two control variables—urbanisation level and FDI—with material capital level and outward FDI (OFDI). We find clear differences in the size of regression coefficients, but no significant change in the sign and significance (see Table 7), confirming robustness.

Table 7
Robustness test.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>	<i>GVCposition</i>
<i>lnTRAN</i>	1.195*** (0.000)	1.241*** (0.000)	1.249*** (0.000)	1.274*** (0.000)	1.277*** (0.000)	1.216*** (0.000)	1.215*** (0.000)
<i>lnPGDP</i>		2.734*** (0.000)	2.554*** (0.000)	2.216*** (0.000)	2.240*** (0.000)	2.222*** (0.000)	2.224*** (0.000)
<i>DE</i>			0.405* (0.013)	0.349* (0.036)	0.348* (0.037)	0.113 (0.511)	0.114 (0.507)
<i>lnCERR</i>				1.039 (0.055)	1.045 (0.055)	−0.195 (0.746)	−0.188 (0.754)
<i>lnGFC</i>					−0.088 (0.914)	−0.389 (0.632)	−0.395 (0.627)
<i>lnCIT</i>						12.364*** (0.000)	12.360*** (0.000)
<i>lnOFDIC</i>							−0.008 (0.837)
_cons	−15.892*** (0.000)	−40.387*** (0.000)	−39.090*** (0.000)	−39.581*** (0.000)	−39.561*** (0.000)	−85.280*** (0.000)	−85.214*** (0.000)
<i>N</i>	1416	1416	1416	1416	1416	1416	1416
adj. R^2	−0.017	0.008	0.012	0.014	0.013	0.029	0.028

p -values in parentheses.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

4.2.5. Transmission mechanism

It is of great practical significance to study the level of the logistics industry for the transmission mechanism of GVC position. If we can find a way to influence the level of the logistics industry on the GVC, we can put forward countermeasures and suggestions for the promotion of China's position in the GVC. We thus examined the effect of the logistics industry level on the GVC status of 59 sample countries through national economic development level, human capital, technological innovation and application level, and logistics infrastructure level. We explored the transmission mechanism of the level of the logistics industry affecting the position of GVC from multiple angles. First, we constructed a transmission mechanism model:

$$\ln Y_{it} = \lambda_0 + \lambda_1 \ln \text{TRAN} + \lambda_2 X_2 + \varepsilon_{it},$$

where Y_{it} represents the set of conduction variables and X_2 represents a series of control variables. The control variables are physical capital level, digital economy level, and FDI. These control variables were closely related to national economic development level, human capital, technological innovation and application level, and logistics infrastructure level. We describe them below:

Outward foreign direct investment (Xing et al., 2021). OFDI is an international investment in the form of diversification. Expanding the scale of FDI helps countries seek technological investment and strengthen the national economy. It can better leverage the catalytic role of GVCs.

Material capital level. A weak market effect will lead to a large number of bad investments that harm economic growth. Thus, the level of a country's physical capital will affect the position of GVC. Following the literature (Su et al., 2020), we measure the level of physical capital by the proportion of fixed capital formation in GDP.

Digital economy level. Based on the perspective of digital industrialisation, Gao and Wang (2020) drew on the definition of digital economy that best suited their objective. We thus looked to the added value of D61 (Telecommunications) and D62T63 (IT and other information services) industries in the OECD TiVA database, the sum of which yields the size of the digital economy. We then divided the size of the digital economy by the economy's total exports to obtain a measure of the level of progress in the digital economy.

Foreign direct investment. Through the introduction of foreign capital, the state alleviates underinvestment, expands employment, develops technology-oriented investment, promotes the export of high-tech products, and promotes economic growth.

The level of logistics industry is taken as the core explanatory variable. We performed a regression of the explained variables $Y = \{PGDP, CER, Inno, INT\}$ and control variables $OFDI, CFCF, DE$, and FDI .

When the explained variables are national economic development level, human capital, technological innovation and application level, and logistics infrastructure level, the logistics industry level is significant at the level of 0.1% (see Table 8). Thus, the level of logistics industry has a significant promoting effect on the level of national economic development, human capital, technological innovation and application, and the level of logistics infrastructure.

We explain the analysis of the logistics level on the mechanism of conduction factors as follows. First, the level of logistics industry promotes the status of GVC by improving the level of national economic development. The logistic level regression coefficient was 0.193, passing the significance test of 0.1% (column 1, Table 6). It shows that the improvement of national logistics industry level can promote economic development. The higher the per capita GDP of a country, the more developed its economy, and the easier it is for a country or region to move up the GVC.

Second, the level of the logistics industry promotes the status of GVC by improving the accumulation of human capital. The logistic level regression coefficient was 0.065, passing the significance test of 0.1% (column 2, Table 6). Thus, the improvement of the level of national logistics industry can promote the accumulation of human capital. The more human capital a country has, the better its ability to learn and innovate, and the easier it is to improve the productivity of enterprises, and thus promote the status of a country or region in the GVC.

Third, the level of the logistics industry promotes the status of GVC by improving the level of technological innovation and application. The logistic level regression coefficient was 0.246, passing the significance test of 0.1% (column 3, Table 6). It shows that the improvement of national logistics industry level can promote the development of technological innovation and application. The innovation and application of technology promote the export of more high-tech products, and the easier it is for a country or region to move up the GVC.

Fourth, the level of the logistics industry promotes the status of GVC by improving the level of logistics infrastructure. The logistic level regression coefficient was 0.568, passing the significance test of 0.1% (column 4, Table 6). It shows that the improvement of national logistics industry level can promote the construction of logistics infrastructure. The better the logistics infrastructure is, the bigger the cargo carrying capacity and export volume will be, and the easier it is for a country or region to move up the GVC.

In summary, the level of the logistics industry affects the promotion of GVC status through various related factors. According to the regression results in Table 4, the level of the logistics industry had a significant and positive influence on the position of GVC. In addition, the logistics industry exerted a transmission effect on the position of GVC through the level of national economic development, human capital, technological innovation and application, and logistics infrastructure.

4.2.6. Result analysis

In this study, we selected the logistics industry statistics of 59 countries from 1995 to 2018. A bidirectional fixed model was constructed to explore the influencing factors of the development of the logistics industry on the position of GVC. The results are outlined below.

First, the logistics industry level has a significant influence on the promotion of a country's position in the GVC. The level of the logistics industry has a more significant effect on the promotion of the GVC position in developing countries. According to the results of

Table 8

The regression results of the transmission mechanism.

Variable	(1)	(2)	(3)	(4)
	<i>lnPGDP</i>	<i>lnCERR</i>	<i>lnInn</i>	<i>lnINTS</i>
<i>lnTRAN</i>	0.193*** (0.000)	0.065*** (0.000)	0.246*** (0.000)	0.568*** (0.000)
<i>lnOFDIC</i>	−0.011 (0.281)	−0.005 (0.364)	0.037 (0.124)	−0.052 (0.096)
<i>lnGFC</i>	0.081 (0.592)	0.196* (0.023)	3.672*** (0.000)	1.138* (0.012)
<i>DE</i>	−0.146*** (0.000)	−0.029* (0.019)	−0.156** (0.002)	0.471*** (0.000)
<i>FDIA</i>	0.479*** (0.000)	0.189*** (0.001)	0.428 (0.057)	0.004 (0.988)
_cons	7.660*** (0.000)	2.659*** (0.000)	8.230*** (0.000)	3.336* (0.027)
<i>N</i>	1416	1416	1416	1416
adj. <i>R</i> ²	0.118	0.040	0.094	0.099

p-values in parentheses.**p* < 0.05.***p* < 0.01.****p* < 0.001.

model (1), the regression coefficient of the logistics industry level was 1.195. After adding six control variables, all passed the significance test at 0.1%. This indicates that an improvement in a country's logistics industry level can improve the promotion of GVC status. The higher the total exports of logistics, the higher the value-added trade. More frequent international trade assists in promoting a country's position in the GVC. The logistics industry and deep integration of Internet digital technology have derived a variety of new commercial logistics business forms. The integration of information technology improves the logistics operation process and innovates the logistics operation mode. It promotes information sharing, reduces operating costs, and improves the efficiency and quality of industry operations. The resource integration of domestic and foreign supply chains is accelerating and will form a new ecology of cross-border intelligent logistics. The continuous expansion and quality improvement of domestic and foreign consumer markets will help promote the high-quality development of the logistics industry. China's participation in the GVC is deepening, and its high-tech, mechanical, and electrical products are expanding overseas markets. The joint construction of the Belt and Road provides China's cross-border e-commerce enterprises with the opportunity to quickly establish the Belt and Road countries' markets. Chinese companies are connecting more overseas markets by 'going global', and the development of China's logistics industry has great globalisation potential, which has a positive effect on the promotion of GVC status.

Second, through the improvement of the national economic development level, human capital, technological innovation, application level, and logistics infrastructure level, the logistics industry showed a significant effect on the promotion of a country's position in the GVC. What we can induce from this result is that enhancing China's core competitiveness requires continuous technological innovation and upgrading. It must improve and upgrade the existing logistics infrastructure and reduce logistics costs to get rid of the 'low-end lock'. Technology- and service-oriented enterprises actively promote the transformation and application of industrial Internet scientific and technological achievements in the rural logistics industry. The construction of a high-quality intelligent logistics system will guide Chinese industries to move to the middle and high-end levels of GVC faster. We need to fully mobilise the vitality of domestic and international demand to boost effective demand and economic growth.

5. Conclusion and policy recommendations

Based on the above analysis, the following conclusions can be drawn, and policy suggestions are provided regarding China's actual situation. The development of the logistics industry plays a positive role in promoting the GVC position, and the continuous growth in total logistics exports can promote the position of a country in the GVC. The level of the logistics industry has a more significant impact on the promotion of GVC position in developing countries. In addition, through the improvement of national economic development level, human capital, technological innovation and application level and logistics infrastructure level, logistics industry has a significant impact on the promotion of a country's position in the GVC.

First, China should promote the construction of logistics infrastructure, and adhere to the principles of market leadership, enterprise operation, code of conduct, and creating a supporting environment. It must employ information technology and reduce logistics costs, improve comprehensive service quality, actively introduce advanced international logistics management experience and technology, increase the innovation of logistics organisation forms, and promote industrial upgrading and consumption upgrading. China should integrate existing logistics resources, accelerate the transformation from traditional logistics to modern logistics, build modern logistics infrastructure platforms and information platforms, cultivate modern logistics parks, develop modern logistics backbone enterprises, build a modern logistics system with complete functions, including transportation, warehousing, distribution, circulation and processing, and improve the service level of the modern logistics industry. Gradually, the logistics industry has become a new growth point and pillar for local economic development.

Second, the development of the modern logistics industry should be promoted, the upstream and downstream division of the logistics supply chain labour cooperation system should be improved, cross-border integration of the logistics industry should be promoted, the construction of export logistics infrastructure and cross-border intelligent logistics systems should be accelerated, and the customs clearance and logistics hub capacity of China should be enhanced. It should actively promote the mutual recognition of international transport road standards, service facilities, traffic signs, and joint inspection results and realise the interoperability of entry and exit means of transport; strengthen multilateral cross-border trade cooperation based on the Belt and Road Initiative, encourage Chinese enterprises to participate in international competition through ‘going out’ and ‘bringing in’; and enhance their export-oriented learning and competitiveness. By increasing the international market share of high-tech products and strengthening the protection of intellectual property rights, we can promote the continuous improvement of new retail, scene exchange, and supply chain finance in the logistics industry. Simultaneously, we will accelerate the improvement of industry standards, guide the new normal of intelligent logistics, and promote the logistics industry to raise the GVC through introduction, absorption, digestion, and re-innovation.

Finally, in the context of China’s reality with the process of building a new development pattern, China’s logistics industry should optimise the chain integration force through “reinforcing chain, strengthening chain and extending chain” to ensure the smooth circulation of the domestic logistics industry, then promote the stable growth of China’s GDP. China must quickly popularise the concept of green sharing and lead to high-quality development of the logistics industry. We should attach importance to cooperation with developed countries, strengthen cooperation with Belt and Road countries and ASEAN countries, and continue to explore potential markets in Africa and Latin America. Promote the logistics industry to expand to the global scope, expand the process of opening up, obtain high value-added in the GVC, and improve the impact and voice of China’s logistics industry in the GVC. At the same time, it will accelerate the upgrading and transformation of the logistics industry; promote the development of new technologies, new business forms, and new models; promote the deep integration of manufacturing and logistics industries; enhance the comprehensive strength of various industries; and promote the status of GVC in the high-quality development stage of China’s economy.

Declaration of competing interest

The authors declare that they have no conflicts of interest.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by National Social Science Foundation Project of China (Research on the realization mechanism and path of value co-creation in digital innovation-driven platform-based enterprises, Grant no.22FGLB041), Social Science Foundation Project of Hunan Province (Spatiotemporal evolution and mechanism of logistics industry efficiency from the perspective of total factor productivity, Grant no.22JD045), Scientific research Project of Education Department of Hunan Province (Research on the realization mechanism and path of value co-creation in digital innovation-driven platform-based enterprises, Grant no.21A0383) and Jiangsu University Philosophy and Social Science Research Project “Research on the Price Discovery of Soybean Futures and Spot Market under the New Situation of International Trade” (2020SJA1387).

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